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・積和の公式を利用した定積分

(例1) 次の定積分を求めよ。

(1) $\int_0^{\pi} \sin 2x \cos 3x dx$ (2) $\int_{\frac{\pi}{8}}^{\frac{\pi}{2}} \cos x \cos 3x dx$

point

$$\circ \sin \alpha \cos \beta = \frac{1}{2} \{ \sin(\alpha + \beta) + \sin(\alpha - \beta) \}$$

$$\triangle \cos \alpha \sin \beta = \frac{1}{2} \{ \sin(\alpha + \beta) - \sin(\alpha - \beta) \}$$

$$\circ \cos \alpha \cos \beta = \frac{1}{2} \{ \cos(\alpha + \beta) + \cos(\alpha - \beta) \}$$

$$\circ \sin \alpha \sin \beta = -\frac{1}{2} \{ \cos(\alpha + \beta) - \cos(\alpha - \beta) \}$$

$$\begin{aligned} (1) \int_0^{\pi} \sin 2x \cos 3x dx &= \int_0^{\pi} \cos 3x \sin 2x dx \\ &= \frac{1}{2} \int_0^{\pi} \{ \sin(3x + 2x) - \sin(3x - 2x) \} dx \\ &= \frac{1}{2} \int_0^{\pi} (\sin 5x - \sin x) dx \\ &= \frac{1}{2} \left[-\frac{1}{5} \cos 5x + \cos x \right]_0^{\pi} \\ &= \frac{1}{2} \left\{ \left(-\frac{1}{5} - 1 \right) - \left(-\frac{1}{5} + 1 \right) \right\} \\ &= -\frac{4}{5} \end{aligned}$$

$$\begin{aligned} (2) \int_{\frac{\pi}{8}}^{\frac{\pi}{2}} \cos x \cos 3x dx &= \int_{\frac{\pi}{8}}^{\frac{\pi}{2}} \cos 3x \cos x dx \\ &= \frac{1}{2} \int_{\frac{\pi}{8}}^{\frac{\pi}{2}} \{ \cos(3x + x) + \cos(3x - x) \} dx \\ &= \frac{1}{2} \int_{\frac{\pi}{8}}^{\frac{\pi}{2}} (\cos 4x + \cos 2x) dx \\ &= \frac{1}{2} \left[\frac{1}{4} \sin 4x + \frac{1}{2} \sin 2x \right]_{\frac{\pi}{8}}^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left\{ 0 - \left(-\frac{\sqrt{2}}{8} + \frac{\sqrt{2}}{4} \right) \right\} \\ &= -\frac{\sqrt{2}}{16} \end{aligned}$$

(例2) m, n は正の整数とする。次の定積分を求めよ。

(1) $\int_0^{\pi} \cos m x \cos n x dx$ (2) $\int_0^{\pi} \sin m x \sin n x dx$ (3) $\int_0^{\pi} \sin m x \cos n x dx$

(1) $\cos m x \cos n x = \frac{1}{2} \{ \cos(m+n)x + \cos(m-n)x \}$ ← 積分して $\frac{1}{m-n}$ がでてくる
 $\neq 0$ はマズイ!

(i) $m \neq n$ のとき

$$\begin{aligned} (\text{与式}) &= \frac{1}{2} \left[\frac{1}{m+n} \sin(m+n)x + \frac{1}{m-n} \sin(m-n)x \right]_0^{\pi} \\ &= 0 \end{aligned}$$

(ii) $m = n$ のとき

$$\begin{aligned} (\text{与式}) &= \frac{1}{2} \int_0^{\pi} (\cos 2mx + 1) dx \\ &= \frac{1}{2} \left[\frac{1}{2m} \sin 2mx + x \right]_0^{\pi} \\ &= \frac{\pi}{2} \end{aligned}$$

よって

$$\int_0^{\pi} \cos m x \cos n x dx = \begin{cases} 0 & (m \neq n) \\ \frac{\pi}{2} & (m = n) \end{cases}$$

(2) $\sin m x \sin n x = -\frac{1}{2} \{ \cos(m+n)x - \cos(m-n)x \}$ ← 積分して $\frac{1}{m-n}$ がでてくる
 $\neq 0$ はマズイ!

(i) $m \neq n$ のとき

$$\begin{aligned} (\text{与式}) &= -\frac{1}{2} \left[\frac{1}{m+n} \sin(m+n)x - \frac{1}{m-n} \sin(m-n)x \right]_0^{\pi} \\ &= 0 \end{aligned}$$

(ii) $m = n$ のとき

$$\begin{aligned} (\text{与式}) &= -\frac{1}{2} \int_0^{\pi} (\cos 2mx - 1) dx \\ &= -\frac{1}{2} \left[\frac{1}{2m} \sin 2mx - x \right]_0^{\pi} \\ &= \frac{\pi}{2} \end{aligned}$$

よって

$$\int_0^{\pi} \sin m x \sin n x dx = \begin{cases} 0 & (m \neq n) \\ \frac{\pi}{2} & (m = n) \end{cases}$$

(3) $\sin m x \cos n x = \frac{1}{2} \{ \sin(m+n)x + \sin(m-n)x \}$ ← 積分して $\frac{1}{m-n}$ がでてくる
 $\neq 0$ はマズイ!

(i) $m \neq n$ のとき

$$(\text{与式}) = -\frac{1}{2} \left[\frac{1}{m+n} \cos(m+n)x + \frac{1}{m-n} \cos(m-n)x \right]_0^{\pi}$$

ここで、 $m+n$ が偶数のとき、 $m-n$ も偶数であるから

$$\begin{aligned} (\text{与式}) &= -\frac{1}{2} \left\{ \left(\frac{1}{m+n} + \frac{1}{m-n} \right) - \left(\frac{1}{m+n} + \frac{1}{m-n} \right) \right\} \\ &= 0 \end{aligned}$$

また、 $m+n$ が奇数のとき、 $m-n$ も奇数であるから

$$\begin{aligned} (\text{与式}) &= -\frac{1}{2} \left\{ \left(-\frac{1}{m+n} - \frac{1}{m-n} \right) - \left(\frac{1}{m+n} + \frac{1}{m-n} \right) \right\} \\ &= \frac{1}{m+n} + \frac{1}{m-n} \\ &= \frac{2m}{m^2 - n^2} \end{aligned}$$

(ii) $m = n$ のとき

$$\begin{aligned} (\text{与式}) &= \frac{1}{2} \int_0^{\pi} \sin 2mx dx \\ &= \frac{1}{2} \left[-\frac{1}{2m} \cos 2mx \right]_0^{\pi} \\ &= 0 \end{aligned}$$

これは、 $m+n$ が偶数のときに含まれる。

よって

$$\int_0^{\pi} \sin m x \cos n x dx = \begin{cases} 0 & (m+n \text{ が偶数}) \\ \frac{2m}{m^2 - n^2} & (m+n \text{ が奇数}) \end{cases}$$